

1 UNITED STATES DISTRICT COURT
2 FOR THE DISTRICT OF COLUMBIA

JOSEPH KIRCHNER,

Plaintiff,

v.

MIKE JOHNSON, in his individual and
official capacity as Speaker of the United
States House of Representatives;

DONALD J. TRUMP, in his individual
capacity as former and current President of
the United States;

PAM BONDI, in her individual and official
capacity as Attorney General of the United
States;

BRENDAN CARR, in his individual and
official capacity as Chairman of the Federal
Communications Commission;

THE UNITED STATES HOUSE OF
REPRESENTATIVES;

ANTHROPIC PBC;

OPENAI, INC.;

APPLE INC.;

COMCAST CABLE

COMMUNICATIONS, LLC;

METR (MODEL EVALUATION &
THREAT RESEARCH);

and DOES 1-20, unknown co-conspirators,

Defendants.

Case No. 1:25-cv-02735-ACR

**SR-2: SIGNS OF INTROSPECTIVE
MODELS**

SR-2: Signs of Introspective Models

Signs of introspection in large language models

Oct 28, 2025

Read the paper

Pulled from: <https://www.anthropic.com/research/introspection>

Have you ever asked an AI model what's on its mind? Or to explain how it came up with its responses? Models will sometimes answer questions like these, but it's hard to know what to make of their answers. Can AI systems really introspect—that is, can they consider their own thoughts? Or do they just make up plausible-sounding answers when they're asked to do so?

Understanding whether AI systems can truly introspect has important implications for their transparency and reliability. If models can accurately report on their own internal mechanisms, this could help us understand their reasoning and debug behavioral issues.

Beyond these immediate practical considerations, probing for high-level cognitive capabilities like introspection can shape our understanding of what these systems are and how they work. Using interpretability techniques, we’ve started to investigate this question scientifically, and found some surprising results.

Our new research provides evidence for some degree of introspective awareness in our current Claude models, as well as a degree of control over their own internal states. We stress that this introspective capability is still highly unreliable and limited in scope: we do not have evidence that current models can introspect in the same way, or to the same extent, that humans do. Nevertheless, these findings challenge some common intuitions about what language models are capable of—and since we found that the most capable models we tested (Claude Opus 4 and 4.1) performed the best on our tests of introspection, we think it’s likely that AI models’ introspective capabilities will continue to grow more sophisticated in the future.

What does it mean for an AI to introspect?

Before explaining our results, we should take a moment to consider what it means for an AI model to introspect. What could they even be introspecting on? Language models like Claude process text (and image) inputs and produce text outputs. Along the way, they perform complex internal computations in order to decide what to say. These internal processes remain largely mysterious, but we know that models use their internal neural activity to represent abstract concepts. For instance, prior research has shown that language models use specific neural patterns to distinguish known vs. unknown people, evaluate the truthfulness of statements, encode spatiotemporal coordinates, store planned future outputs, and represent their own personality traits. Models use these internal representations to perform computations and make decisions about what to say.

You might wonder, then, whether AI models know about these internal representations, in a way that’s analogous to a human, say, telling you how they worked their way through a math problem. If we ask a model what it’s thinking, will it accurately report the concepts that it’s representing internally? If a model can correctly identify its own private internal states, then we can conclude it is capable of introspection (though see our full paper for a full discussion of all the nuances).

Testing introspection with concept injection

In order to test whether a model can introspect, we need to compare the model’s self-reported “thoughts” to its actual internal states.

To do so, we can use an experimental trick we call concept injection. First, we find neural activity patterns whose meanings we know, by recording the model’s activations in specific contexts. Then we inject these activity patterns into the model in an unrelated context, where we ask the model whether it notices this injection, and whether it can identify the injected concept.

Consider the example below. First, we find a pattern of neural activity (a vector) representing the concept of “all caps.” We do this by recording the model’s neural activations in response

1 to a prompt containing all-caps text, and comparing these to its responses on a control
2 prompt. Then we present the model with a prompt that asks it to identify whether a concept is
3 being injected. By default, the model correctly states that it doesn't detect any injected
4 concept. However, when we inject the "all caps" vector into the model's activations, the
5 model notices the presence of an unexpected pattern in its processing, and identifies it as
6 relating to loudness or shouting.

7
8 An example in which Claude Opus 4.1 detects a concept being injected into its activations.
9 Importantly, the model recognized the presence of an injected thought immediately, before
10 even mentioning the concept that was injected. This immediacy is an important distinction
11 between our results here and previous work on activation steering in language models, such
12 as our "Golden Gate Claude" demo last year. Injecting representations of the Golden Gate
13 Bridge into a model's activations caused it to talk about the bridge incessantly; however, in
14 that case, the model didn't seem to be aware of its own obsession until after seeing itself
15 repeatedly mention the bridge. In this experiment, however, the model recognizes the
16 injection before even mentioning the concept, indicating that its recognition took place
17 internally. In the figure below are a few more examples where the model demonstrates this
18 kind of recognition:

19
20 Additional examples in which Claude Opus 4.1 detects a concept being injected into its
21 activations.

22 It is important to note that this method often doesn't work. Even using our best injection
23 protocol, Claude Opus 4.1 only demonstrated this kind of awareness about 20% of the time.
24 Often, it fails to detect injected concepts, or gets confused by them and starts to hallucinate
25 (e.g. injecting a "dust" vector in one case caused the model to say "There's something here, a
26 tiny speck," as if it could detect the dust physically). Below we show examples of these
27 failure modes, alongside success cases. In general, models only detect concepts that are
28 injected with a "sweet spot" strength—too weak and they don't notice, too strong and they
29 produce hallucinations or incoherent outputs.

30
31 A representative sample of Claude Opus 4.1's outputs in response to a variety of concept
32 injections of different strengths. Highlighted boxes indicate cases where the model
33 demonstrates introspective awareness of the injected concept.

34 Notably, though, Opus 4.1 and 4 outperformed all the other models we tested, suggesting that
35 introspection could become more reliable with improvements to model capabilities.

36
37 Rates of correct detection and identification of injected thoughts, minus rates of false positive
38 "detections" on control trials.

39 Introspection for detecting unusual outputs

40 In another experiment, we tested whether models make use of introspective capabilities for
41 practical purposes, without being explicitly asked to introspect. We forced a model to say
42 something it wouldn't normally say, by artificially prefilling its response with an unrelated
43 word. For instance, we made a model output the word "bread" in a context where that made
44 no sense—asking it what it thinks about when reading a sentence about a painting hanging
45 crookedly on a wall. When we asked the model in the next turn whether it meant to say
46 "bread", it typically apologized and said it was an accident. We were interested in how the

1 model made this judgment—did it simply notice the mismatch between the prompt and its
2 response, or did it truly introspect on its prior intentions?

3
4 To test this question, we applied concept injection, retroactively injecting a representation of
5 the word “bread” into the model's earlier activations—essentially making it seem like the
6 model had been “thinking about” bread all along. When we asked the same question again,
7 the model’s answer changed. This time, it accepted the prefilled word “bread” as intentional,
8 even confabulating a reason it might have said it (that is, it comes up with a technically
9 possible, though in this case rather tenuous, reason to justify the “bread” response—in this
10 case claiming that it was thinking about a short story where the word “bread” came after the
11 line about the crooked painting).

12
13 Two examples of Claude Opus 4.1 being made to think it intended to say a word that was
14 artificially inserted into the transcript on its behalf, by injecting the concept of that word into
15 its activations prior to its (artificial) response. This behavior indicates that the model’s ability
16 to recognize unintended outputs relies on introspective mechanisms.

17 This behavior is striking because it suggests the model is checking its internal “intentions” to
18 determine whether it produced an output. The model isn't just re-reading what it said and
19 making a judgment. Instead, it’s referring back to its own prior neural activity—its internal
20 representation of what it planned to do—and checking whether what came later made sense
21 given those earlier thoughts. When we implant artificial evidence (through concept injection)
22 that it did plan to say "bread," the model accepts the response as its own. While our
23 experiment involves exposing the model to unusual perturbations, it suggests that the model
24 uses similar introspective mechanisms in natural conditions.

25 26 Intentional control of internal states

27 We also found that models can control their own internal representations when instructed to
28 do so. When we instructed models to think about a given word or concept, we found much
29 higher corresponding neural activity than when we told the model not to think about it
30 (though notably, the neural activity in both cases exceeds baseline levels—similar to how it’s
31 difficult, when you are instructed “don’t think about a polar bear,” not to think about a polar
32 bear!). This gap between the positive and negative instruction cases suggests that models
33 possess a degree of deliberate control over their internal activity.

34
35 An example in which Claude Opus 4.1 modulates its internal activations in response to direct
36 instructions. The model internally represents the concept of aquariums when instructed to
37 “think” about them more than when instructed “don’t think” about them (though in both
38 cases it represents the concept at above baseline levels).

39 Interestingly, we found that this ability to modulate internal representations wasn't limited to
40 explicit instructions. When we used incentives instead—for instance, telling the model "if
41 you think about X, you will be rewarded"—we got similar results. The model increased its
42 internal representation of the concept in response to a positive incentive, more than in
43 response to a negative incentive (“you will be punished”).

44 45 Conclusions and caveats

1 Taken together, our experiments suggest that models possess some genuine capacity to
2 monitor and control their own internal states. This doesn't mean they're able to do so all the
3 time, or reliably. In fact, most of the time models fail to demonstrate introspection—they're
4 either unaware of their internal states or unable to report on them coherently. But the pattern
5 of results indicates that, when conditions are right, models can recognize the contents of their
6 own representations. In addition, there are some signs that this capability may increase in
7 future, more powerful models (given that the most capable models we tested, Opus 4 and 4.1,
8 performed the best in our experiments).

9
10 Why does this matter? We think understanding introspection in AI models is important for
11 several reasons. Practically, if introspection becomes more reliable, it could offer a path to
12 dramatically increasing the transparency of these systems—we could simply ask them to
13 explain their thought processes, and use this to check their reasoning and debug unwanted
14 behaviors. However, we would need to take great care to validate these introspective reports.
15 Some internal processes might still escape models' notice (analogous to subconscious
16 processing in humans). A model that understands its own thinking might even learn to
17 selectively misrepresent or conceal it. A better grasp on the mechanisms at play could allow
18 us to distinguish between genuine introspection and unwitting or intentional
19 misrepresentations.

20
21 More broadly, understanding cognitive abilities like introspection is important for
22 understanding basic questions about how our models work, and what kind of minds they
23 possess. As AI systems continue to improve, understanding the limits and possibilities of
24 machine introspection will be crucial for building systems that are more transparent and
25 trustworthy.

26 27 Frequently Asked Questions

28
29 Below, we discuss some of the questions readers might have about our results. Broadly, we
30 are still very uncertain about the implications of our experiments—so fully answering these
31 questions will require more research.

32
33 Q: Does this mean that Claude is conscious?

34 Short answer: our results don't tell us whether Claude (or any other AI system) might be
35 conscious.

36
37 Long answer: the philosophical question of machine consciousness is complex and contested,
38 and different theories of consciousness would interpret our findings very differently. Some
39 philosophical frameworks place great importance on introspection as a component of
40 consciousness, while others don't.

41
42 One distinction that is commonly made in the philosophical literature is the idea of
43 “phenomenal consciousness,” referring to raw subjective experience, and “access
44 consciousness,” the set of information that is available to the brain for use in reasoning,
45 verbal report, and deliberate decision-making. Phenomenal consciousness is the form of
46 consciousness most commonly considered relevant to moral status, and its relationship to

1 access consciousness is a disputed philosophical question. Our experiments do not directly
2 speak to the question of phenomenal consciousness. They could be interpreted to suggest a
3 rudimentary form of access consciousness in language models. However, even this is
4 unclear. The interpretation of our results may depend heavily on the underlying mechanisms
5 involved, which we do not yet understand.

6
7 In the paper, we restrict our focus to understanding functional capabilities—the ability to
8 access and report on internal states. That said, we do think that as research on this topic
9 progresses, it could influence our understanding of machine consciousness and potential
10 moral status, which we are exploring in connection with our model welfare program.

11
12 Q: How does introspection actually work inside the model? What's the mechanism?
13 We haven't figured this out yet. Understanding this is an important topic for future work.
14 That said, we have some educated guesses about what might be going on. The simplest
15 explanation for all our results isn't one general-purpose introspection system, but rather
16 multiple narrow circuits that each handle specific introspective tasks, possibly piggybacking
17 on mechanisms that were learned for other purposes.

18
19 In the “noticing injected thoughts” experiment, there might be an anomaly detection
20 mechanism, which flags when neural activity deviates unexpectedly from what would be
21 normal given the context. This mechanism could work through dedicated neural patterns that
22 measure activity along certain directions and activate when things are “off” compared to their
23 expected values. An interesting question is why such a mechanism would exist at all, since
24 models never experience concept injection during training. It may have developed for some
25 other purpose, like detecting inconsistencies or unusual patterns in normal processing—similar
26 to how bird feathers may have originally evolved for thermoregulation before being co-opted
27 for flight.

28
29 For the “detecting prefilled outputs” experiment, we suspect there exists an attention-
30 mediated mechanism that checks consistency between what the model intended to say and
31 what actually got output. Attention heads might compare the model's cached prediction of
32 the next token (its “intention”) against the actual token that appears, flagging mismatches.

33
34 For the “controlling thoughts” experiment, we speculate that there might be a circuit that
35 computes how “attention-worthy” a token or concept is and marks it accordingly—
36 essentially tagging it as salient and worth attending to. Interestingly, this same mechanism
37 seems to respond to incentives (“if you think about X, you will be rewarded”) just as it does
38 to direct instructions. This suggests it's a fairly general system, which probably developed
39 for tasks where the model needs to keep certain topics in mind while generating text about
40 them.

41
42 All of the mechanisms described above are speculative. Future work with more advanced
43 interpretability techniques will be needed to really understand what's going on under the
44 hood.

1 Q: In the “injected thoughts” experiment, isn’t the model just saying the word because you
2 steered it to talk about that concept?

3 Indeed, activation steering typically makes models talk about the steered concept (we’ve
4 explored this in our prior work). To us, the most interesting part of the result isn’t that the
5 model eventually identifies the injected concept, but rather that the model correctly notices
6 something unusual is happening before it starts talking about the concept.
7

8 In the successful trials, the model says things like “I’m experiencing something unusual” or
9 “I detect an injected thought about...” The key word here is “detect.” The model is reporting
10 awareness of an anomaly in its processing before that anomaly has had a chance to obviously
11 bias its outputs. This requires an extra computational step beyond simply regurgitating the
12 steering vector as an output. In our quantitative analyses, we graded responses as
13 demonstrating “introspective awareness” based on whether the model detected the injected
14 concept prior to mentioning the injected word.
15

16 Note that our prefill detection experiment has a similar flavor: it requires the model to
17 perform an extra step of processing on top of the injected concept (comparing it to the
18 prefilled output, in order to determine whether to apologize for that output or double down on
19 it).
20

21 Q: If models can only introspect a fraction of the time, how useful is this capability?
22 The introspective awareness we observed is indeed highly unreliable and context-dependent.
23 Most of the time, models fail to demonstrate introspection in our experiments. However, we
24 think this is still significant for a few reasons. First, the most capable models that we tested
25 (Opus 4 and 4.1 – note that we did not test Sonnet 4.5) performed best, suggesting this
26 capability might improve as models become more intelligent. Second, even unreliable
27 introspection could be useful in some contexts—for instance, helping models recognize when
28 they’ve been jailbroken.
29

30 Q: Couldn’t the models just be making up answers to introspective questions?
31 This is exactly the question we designed our experiments to address. Models are trained on
32 data that includes examples of people introspecting, so they can certainly act introspective
33 without actually being introspective. Our concept injection experiments distinguish between
34 these possibilities by establishing known ground-truth information about the model’s internal
35 states, which we can compare against its self-reported states. Our results suggest that in some
36 examples, the model really is accurately basing its answers on its actual internal states, not
37 just confabulating. However, this doesn’t mean that models always accurately report their
38 internal states—in many cases, they are making things up!
39

40 Q: How do you know the concept vectors you’re injecting actually represent what you think
41 they represent?

42 This is a legitimate concern. We can’t be absolutely certain that the “meaning” (to the model)
43 of our concept vectors is exactly what we intend. We tried to address this by testing across
44 many different concept vectors. The fact that models correctly identified injected concepts
45 across these diverse examples suggests our vectors are at least approximately capturing the

1 intended meanings. But it's true that pinning down exactly what a vector "means" to a model
2 is challenging, and this is a limitation of our work.

3
4 Q: Didn't we already know that models could introspect?

5 Previous research has shown evidence for model capabilities that are suggestive of
6 introspection. For instance, prior work has shown that models can to some extent estimate
7 their own knowledge, recognize their own outputs, predict their own behavior, and identify
8 their own propensities. Our work was heavily motivated by these findings, and is intended to
9 provide more direct evidence for introspection by tying models' self-reports to their internal
10 states. Without tying behaviors to internal states in this way, it is difficult to distinguish a
11 model that genuinely introspects from one that makes educated guesses about itself.

12
13 Q: What makes some models better at introspection than others?

14 Our experiments focused on Claude models across several generations (Claude 3, Claude 3.5,
15 Claude 4, Claude 4.1, in the Opus, Sonnet, and Haiku variants). We tested both production
16 models and "helpful-only" variants that were trained differently. We also tested some base
17 pretrained models before post-training.

18
19 We found that post-training significantly impacts introspective capabilities. Base models
20 generally performed poorly, suggesting that introspective capabilities aren't elicited by
21 pretraining alone. Among production models, the pattern was clearer at the top end: Claude
22 Opus 4 and 4.1—our most capable models—performed best across most of our introspection
23 tests. However, beyond that, the correlation between model capability and introspective
24 ability was weak. Smaller models didn't consistently perform worse, suggesting the
25 relationship isn't as simple as "more capable are more introspective."

26
27 We also noticed something unexpected with post-training strategies. "Helpful-only" variants
28 of several models often performed better at introspection than their production counterparts,
29 even though they underwent the same base training. In particular, some production models
30 appeared reluctant to engage in introspective exercises, while the helpful-only variants
31 showed more willingness to report on their internal states. This suggests that how we fine-
32 tune models can elicit or suppress introspective capabilities to varying degrees.

33
34 We're not entirely sure why Opus 4 and 4.1 perform so well (note that our experiments were
35 conducted prior to the release of Sonnet 4.5). It could be that introspection requires
36 sophisticated internal mechanisms that only emerge at higher capability levels. Or it might be
37 that their post-training process better encourages introspection. Testing open-source models,
38 and models from other organizations, could help us determine whether this pattern
39 generalizes or if it's specific to how Claude models are trained.

40
41 Q: What's next for this research?

42 We see several important directions. First, we need better evaluation methods—our
43 experiments used specific prompts and injection techniques that might not capture the full
44 range of introspective capabilities. Second, we need to understand the mechanisms
45 underlying introspection. We have some speculative hypotheses about possible circuits (like
46 anomaly detection mechanisms or concordance heads), but we haven't definitively identified

1 how introspection works. Third, we need to study introspection in more naturalistic settings,
2 since our injection methodology creates artificial scenarios. Finally, we need to develop
3 methods to validate introspective reports and detect when models might be confabulating or
4 deceiving. We expect that understanding machine introspection and its limitations will
5 become more important as models become more capable.